

Dog Shelter Management System Software Requirements Specification

Fernando Camberos

Kwane Galang

Korapat Jaengsrivong

Andy Lim

Fredy Lung

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Revision History

Revision #	Change Description	Revision Date	Revised By	Approved By
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1.0	Initial document with system overviews and core features	05/5/2026		
1.1	Added Dog Intake and Status Tracking requirements/design	05/6/2026		
1.2	Added Medical Records and Adoption System	05/7/2026		
1.3	Final Checkpoint & Final updates, refinements, and future work	05/8/2026		

1 Introduction

1.1 Purpose

This document defines the software requirements for the Dog Shelter Management System. The system will help shelter staff manage dog intake, medical records, adoption applications, and dog availability status. This SRS describes what the system must do, who will use it, and what constraints or rules apply.

1.2 Intended Audience

This document is intended for developers, testers, project managers, shelter staff, and instructors reviewing the project.

1.3 Product Scope

The Dog Shelter Management System will provide a centralized web-based system for managing shelter dogs from intake through adoption. The system will allow staff to add dog records, track medical history, update adoption status, and review adoption applications submitted by users.

1.4 Definitions, Acronyms, and Abbreviations

- Applicant: A user who submits an adoption application
- Staff: Shelter employee who manages dog and application records
- Admin: Staff member with full system permissions
- Intake: The process of adding a new dog to the shelter system
- Adoption Status: Current state of a dog: Available, Pending, Adopted, or Medical Hold

2 Overall Description

2.1 System Analysis

Shelters often track dogs, medical records, and adoption applications using paper forms or disconnected files. This can lead to missing records, delayed application reviews, and incorrect availability information. The Dog Shelter Management System solves this by storing all major shelter records in one organized system.

2.2 Product Perspective

The system will function as an independent shelter management application. It may later connect to email services, online adoption portals, or donation systems, but those are not required for the first release.

2.3 Product Functions

The system will provide:

- Dog intake record creation
- Dog profile viewing and editing
- Medical record tracking
- Vaccination tracking
- Adoption application submission
- Staff approval/rejection workflow
- Dog status tracking
- Optional admining dashboard for staff management

2.4 User Classes and Characteristics

Applicant: Public user who views dogs and submits adoption applications

Staff: Shelter worker who manages dogs, records, and applications

Admin: Higher-level staff user who manages all system data and staff access

2.5 Operating Environment

The system will run as a web application accessible through modern browsers such as Chrome, Edge, Safari, or Firefox. Staff and applicants will access the system using desktop or mobile devices.

2.6 Assumptions and Dependencies

- Users have internet access
- Staff accounts are created before staff can manage records.
- Dog records must exist before medical records or applications can be connected to them.
- The shelter is responsible for verifying adoption applicant information outside the system.

3 External Interface Requirements

3.1 User Interfaces

The system shall include pages for dog listings, dog profiles, intake forms, medical record forms, adoption applications, application review, and staff dashboard controls.

3.2 Hardware Interfaces

No special hardware interfaces are required.

3.3 Software Interfaces

The system may use a database to store dog, medical, user, and application records. If email notifications are added, the system may connect to an email service.

3.4 Communications Interfaces

The system shall communicate through standard web protocols such as HTTPS.

4 Requirements Specification

4.1 Functional Requirements

Dog Intake System:

- FR-1: The system shall allow authorized staff to create a new dog profile.
- FR-2: The system shall store each dog's name, breed, estimated age, sex, arrival date, and intake notes.
- FR-3: The system shall assign each new dog an initial status of Available, Pending, Medical Hold, or Unavailable.
- FR-4: The system shall allow authorized staff to edit dog profile information.

Medical Records System:

- FR-5: The system shall allow authorized staff to add medical records to a dog profile.
- FR-6: The system shall store vaccination name, vaccination date, due date, medical notes, and current health status.
- FR-7: The system shall allow staff to update a dog's health status.
- FR-8: The system shall prevent applicants from editing medical records.

Adoption Management System:

- FR-9: The system shall allow users to submit adoption applications for available dogs.
- FR-10: The system shall store applicant name, contact information, selected dog, housing information, pet experience, and application notes.
- FR-11: The system shall allow staff to approve, reject, or mark applications as pending.
- FR-12: The system shall record the staff member and date when an application status is changed.

Dog Status Tracking:

- FR-13: The system shall allow staff to update a dog's adoption status.
- FR-14: The system shall remove adopted dogs from the list of available dogs.
- FR-15: The system shall prevent users from applying for dogs marked Adopted or Medical Hold.
- FR-16: The system shall display each dog's current status on its profile.

Admin Dashboard:

- FR-17: The system should provide a dashboard where staff can view total dogs, available dogs, pending applications, and adopted dogs.
- FR-18: The system should allow admins to manage staff accounts.

4.2 External Interface Requirements

4.2.1 Dog Intake Form

The system shall provide a dog intake form for authorized staff to enter new dog records.

Input Fields:

- Dog name
- Breed
- Estimated age
- Sex
- Arrival date
- Intake reason
- Initial status
- Intake notes

Output:

- A new dog profile shall be created and stored in the system.
- The dog shall appear in the staff dog management view.
- If the dog is marked Available, the dog may appear in the public available dogs list.

Validation:

- Dog name, arrival date, and initial status shall be required.
- Arrival date shall not be later than the current date.

4.2.2 Medical Record Form

The system shall provide a medical record form for authorized staff to add or update medical information for a dog.

Input Fields:

- Dog ID
- Vaccination name
- Vaccination date
- Due date
- Medical notes
- Health status

Output:

- The medical record shall be attached to the selected dog profile.
- The dog's current health status shall be updated when applicable.

Validation:

- Medical records shall be connected to an existing dog.
- Applicants shall not be able to access medical record editing controls.

4.2.3 Adoption Application Form

The system shall provide an adoption application form for applicants.

Input Fields:

- Applicant name
- Contact Information
- Address
- Housing type
- Previous pet experience
- Selected dog
- Application notes

Output:

- A new adoption application shall be stored with Pending status.
- Staff shall be able to view the application in the review queue.

Validation:

- Applications shall only be submitted for dogs marked Available.
- Applicant name, contact information, and selected dog shall be required.

4.2.4 Application Review Interface

The system shall provide an application review interface for staff.

Input Fields/Actions:

- Approve application
- Reject application
- Mark application as Pending
- Add staff review notes

Output:

- The application status shall be updated.
- The selected dog's status shall be updated to Pending or Adopted when applicable.
- The system shall record the review date and the reviewing staff member.

Validation:

- Only authorized staff shall review applications.

4.3 Logical Database Requirements

The system shall store records for:

- Dogs
- Medical records
- Vaccinations
- Adoption applications
- Users/Staff accounts
- Dog status history, if included

Basic Relationships:

- One dog may have many medical records.
- One dog may have many adoption applications.
- One applicant may submit multiple applications.
- Staff users may review multiple applications.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-1: The system shall load dog search results within 3 seconds under normal usage.
- NFR-2: The system shall support multiple staff users accessing records at the same time.

5.2 Safety Requirements

- NFR-3: The system shall allow staff to mark dogs as Medical Hold when they are not ready for adoption.
- NFR-4: The system shall prevent dogs with Medical Hold status from being submitted for adoption.

5.3 Security Requirements

- NFR-5: The system shall require staff users to log in before managing records.
- NFR-6: The system shall restrict applicant personal information to authorized staff.
- NFR-7: The system shall prevent public users from editing dog, medical, or application review records.

5.4 Software Quality Attributes

- The system shall be easy for shelter staff to learn.
- The system shall organize dog and application records clearly.
- The system shall be maintainable so future features, such as foster tracking or donations, can be added.

5.5 Business Rules

- Only staff may add or edit dog records.
- Only staff may update medical records.
- Only staff may approve or reject adoption applications.
- Applicants may only apply for dogs marked Available.
- Adopted dogs shall no longer appear as available.
- Dogs on Medical Hold shall not be available for adoption.

6 Legal and Ethical Considerations

The system will store applicant personal information, so it must protect user privacy and limit access to authorized staff. Medical records must be accurate because they affect animal welfare and adoption decisions. The system should support responsible adoption by preventing dogs with unresolved medical concerns from being adopted until cleared by staff.

Glossary

Acronym	Meaning	Description
SRS	Software Requirements Specification	A document that describes what the software system must do, including functional and nonfunctional requirements.
SRD	Software Requirements Document	Another name sometimes used for a requirements document; in this project, it refers to the SRS-style requirements document.
FR	Functional Requirement	A required system behavior or feature, usually written as “The system shall...”
NFR	Nonfunctional Requirement	A requirement describing system qualities such as performance, security, usability, or reliability.
ID	Identifier	A unique value used to identify a record, such as a Dog ID or application ID.
HTTPS	Hypertext Transfer Protocol Secure	A secure web communication protocol used to protect data sent between users and the system.